Experiment No. 10: Image Rotation by 90 Degrees Clockwise

Aim: To rotate the given image by 90 degrees clockwise using Python and OpenCV.

Program (Google Colab):

```
import cv2
import matplotlib.pyplot as plt
from google.colab import files

# Upload an image
uploaded = files.upload()
image_path = next(iter(uploaded))

# Read the image
img = cv2.imread(image_path)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# Rotate image 90 degrees clockwise
rotated = cv2.rotate(img, cv2.ROTATE_90_CLOCKWISE)

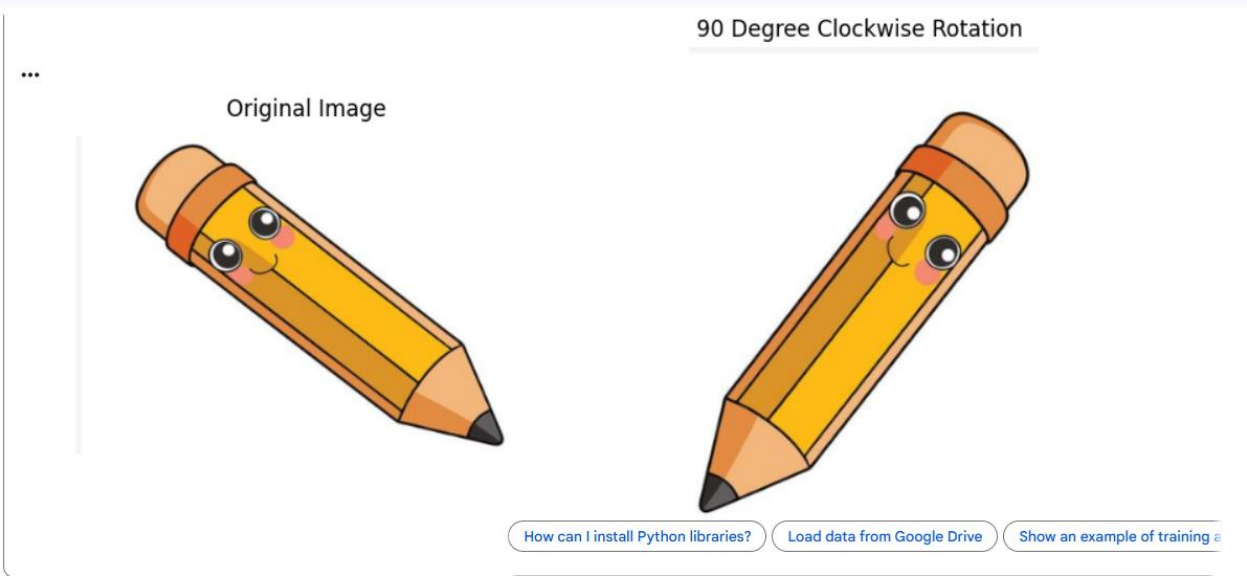
# Display original and rotated images
plt.figure(figsize=(10, 5))

plt.subplot(1, 2, 1)
plt.imshow(img)
plt.title("Original Image")
plt.axis("off")
```

```
plt.subplot(1, 2, 2)
plt.imshow(rotated)
plt.title("90 Degree Clockwise Rotation")
plt.axis("off")

plt.show()
```

OUTPUT:



Expected Output:

1. Original image.
2. Image rotated by 90 degrees clockwise.

Result: The given image was successfully rotated by 90 degrees clockwise using Python and OpenCV's cv2.rotate() function.